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Unit 2 - Bringing Ideas to Life

PDF Documents

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UNIT 2 BRINGING IDEAS TO LIFE

Project Introduction

INTRODUCTION

In this unit, we will learn how to build a Smart Posture Detector. This device will help us monitor our posture and remind us to sit or stand correctly. Using sensors and MODI modules, we will create a tool that can make a big difference in our health and well-being.

IDEA GENERATION

The Smart Posture Detector we are going to build will use sensors to monitor your distance from the monitor and give you feedback when you are too close. Here's how it works:

1. DETECTING POSTURE

The device uses sensors to measure how far your head is from the monitor. It helps ensure you are sitting at a safe distance.

2. PROCESSING INFORMATION

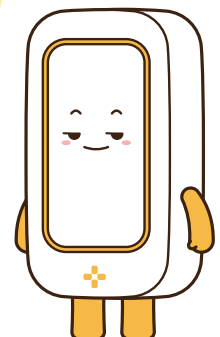
The device uses sensors to measure how far your head is from the monitor. It helps ensure you are sitting at a safe distance.

3. PROVIDING FEEDBACK

The device provides immediate feedback, like a sound or light signal, to alert you when you are too close to the monitor. This helps you adjust your position right away.

TIPS

Think about the key parts of a Smart Posture Detector: sensors to detect distance, a way to process this information, and something to alert you when you are too close to the monitor. What types of MODI modules could you use for each of these parts? Discuss your ideas with your group and write down the modules you think will work best.

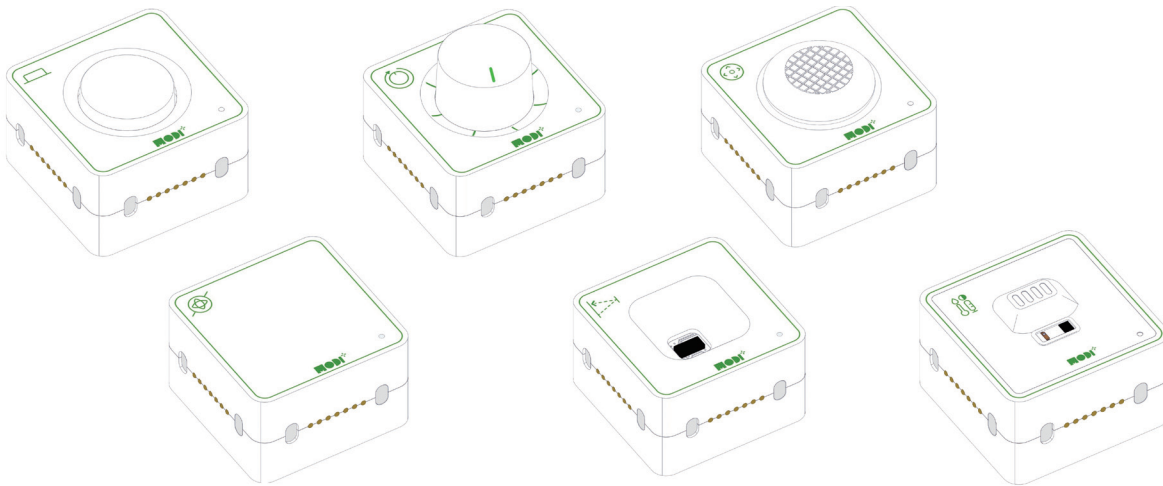


Required Modules & Brainstorming

To build our Smart Posture Detector, we need to decide which MODI modules to use. Here are some ideas to get started:

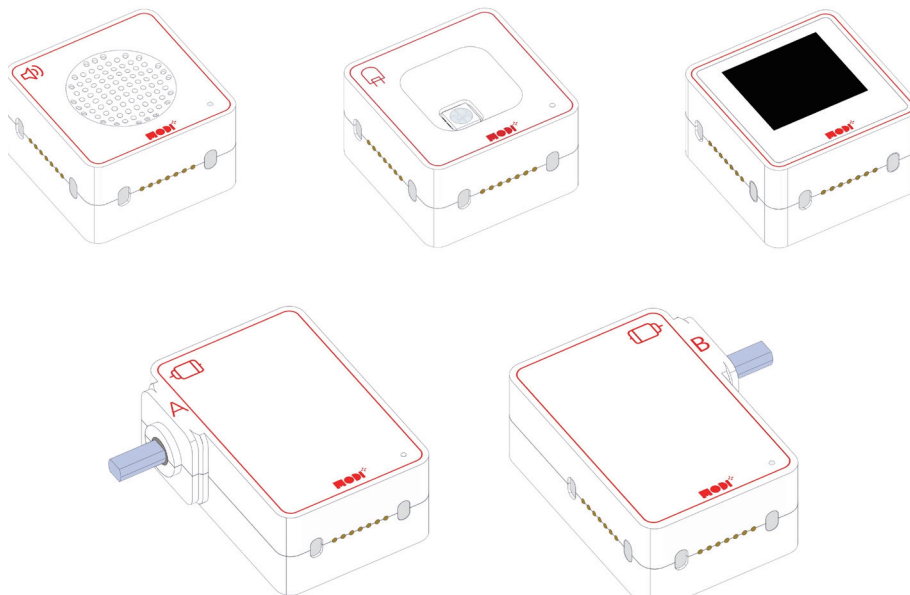
SENSOR MODULES

We need modules that can accurately detect the distance between the user and the monitor. Think about how different types of sensors could help in measuring these distances.



OUTPUT MODULES

These modules will alert you when you are too close to the monitor. Think about how you would like to receive feedback - through sound, light, or another method.



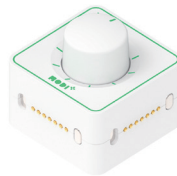


Required Modules & Brainstorming

Which MODI modules do you need for the creation?



BUTTON



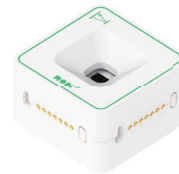
DIAL



ENVIRONMENT



JOYSTICK



TOF



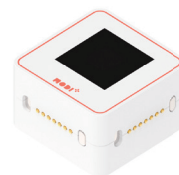
SPEAKER



IMU



LED



DISPLAY

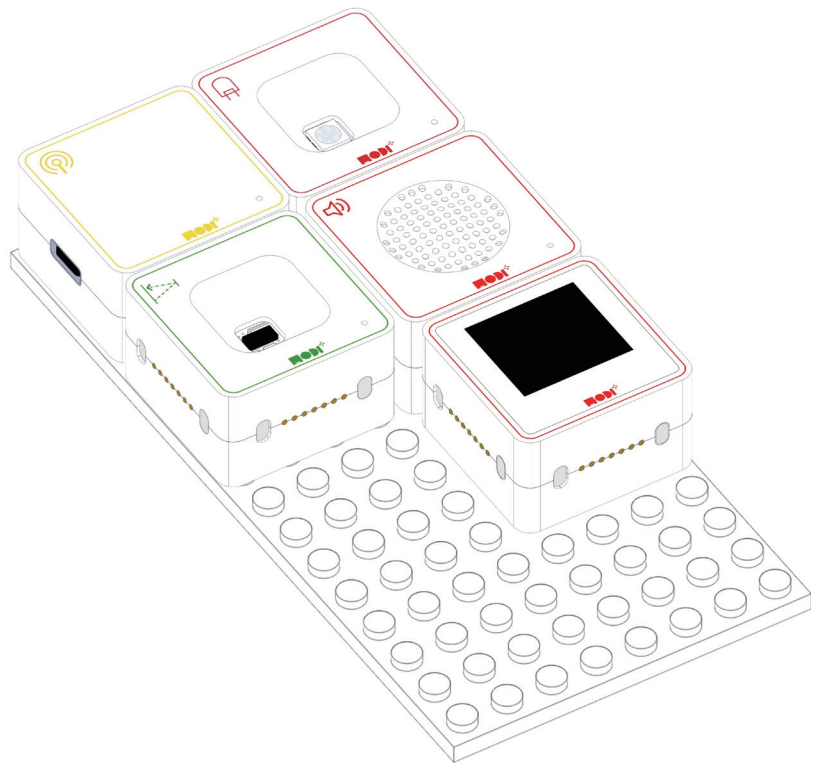
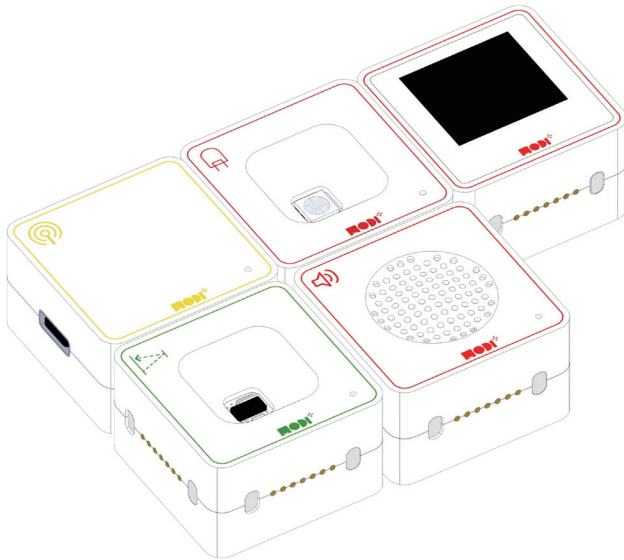


MOTOR A



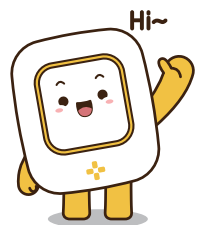
MOTOR B

WHEN WE COMBINE ALL THE MODULES, THE CREATION HAS TO BE LOOK LIKE THIS.



TIPS

MODI modules can be connected in any direction! This makes building your projects easier and more flexible. No need to worry about which way they face!





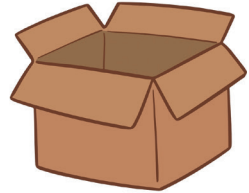
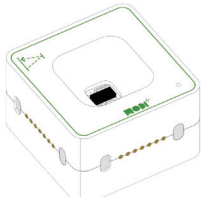
Module Functionality

TOF SENSOR

Time-of-Flight (ToF) Sensor: The ToF sensor measures how far away something is by using light. Here's how it works in simple steps:

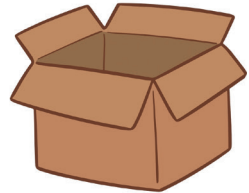
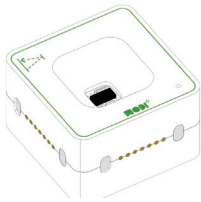
1. SENDING LIGHT

The ToF sensor sends out a small burst of light.



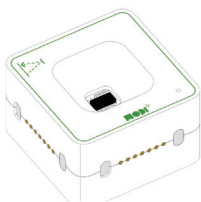
2. BOUNCING BACK

This light travels until it hits an object and then bounces back to the sensor.



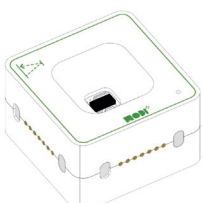
3. TIMING THE JOURNEY

The sensor measures how long it took for the light to go out and come back.

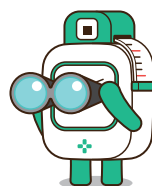


4. CALCULATING DISTANCE

Using the time it took, the sensor calculates the distance to the object. Since light travels really fast, the sensor can figure out the distance very quickly and accurately.



123 cm ...



USES OF THE TOF SENSOR

The ToF sensor is used in many cool ways because it can measure distance very well. Here are some places where you might find it:

SMARTPHONES & TABLETS

ToF sensors help with facial recognition and detecting how close your face or hand is to the device



ROBOTS

ToF sensors help robots move around by measuring how far they are from obstacles. This helps robots avoid bumping into things.



FACTORIES

In factories, ToF sensors measure the distance to make sure parts are in the right place during assembly.



HEALTHCARE

ToF sensors are used in healthcare to monitor patients' movements, helping with things like fall detection and physical therapy.

GAMES & VIRTUAL REALITY

ToF sensors make games more fun by tracking your movements and putting them into the game.

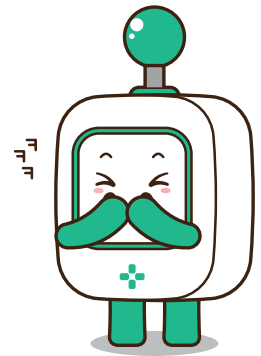




USING TOF SENSORS IN A SMART POSTURE DETECTOR

In our Smart Posture Detector, the ToF sensor will check how far your head and shoulders are from the sensor. By constantly measuring these distances, the sensor helps see if you are sitting or standing up straight. If the sensor notices that you are slouching or hunching, it can send a signal to remind you to sit or stand up straight.

Understanding how the ToF sensor works and where it is used helps us see why it's a great choice for our Smart Posture Detector. This device will help us keep a good posture and stay healthy.



TIPS

Think about the key parts of a Smart Posture Detector: sensors to detect posture, a way to process this information, and something to alert you when your posture needs correcting.

